

CELLARHEAD, United Kingdom

WMO: 033380

Lat: 53.033N

Lon: 2.083W

Elev: 228

StdP: 98.62

Time Zone: 0.00 (EUW)

Period: 86-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.0	-3.6	-7.0	2.1	-3.4	-5.6	2.4	-2.0	12.5	4.3	11.6	6.1	1.2	90	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	24.2	17.2	22.0	16.2	20.3	15.5	17.9	22.4	17.0	21.0	16.1	19.3	3.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.9	19.2	15.4	11.3	18.3	14.5	10.6	17.4	51.1	22.4	48.4	21.2	45.6	19.3	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	8.0	DB	-8.2	26.8	2.8	2.6	-10.3	28.7	-11.9	30.2	-13.5	31.7	-15.5	33.5	(4)
(5)			WB	-8.6	19.6	2.7	1.3	-10.6	20.5	-12.2	21.3	-13.7	22.1	-15.7	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	2.6	2.6	4.5	6.6	9.8	12.5	14.9	14.3	11.5	9.2	5.5	3.7		
	DBStd	5.21	3.12	3.37	2.73	2.93	3.11	2.99	2.56	2.86	2.45	2.83	3.20	3.29		
	HDD10.0	1161	228	206	170	109	42	8	0	1	12	48	139	197		
	HDD18.3	3721	487	439	428	352	265	178	111	131	206	283	387	454		
	CDD10.0	496	0	0	1	7	36	83	152	134	56	24	2	1		
	CDD18.3	12	0	0	0	0	0	2	4	6	0	0	0	0		
	CDH23.3	100	0	0	0	0	0	12	37	51	0	0	0	0		
	CDH26.7	14	0	0	0	0	0	1	2	11	0	0	0	0		
(14) Wind	WSAvg	3.5	4.1	4.3	4.6	3.6	3.5	3.0	3.1	2.9	2.9	3.5	3.4	3.4		
(15) Precipitation	PrecAvg	978	90	71	64	64	69	76	78	86	80	102	98	102		
	PrecMax	1284	170	167	128	130	123	173	171	230	136	200	200	189		
	PrecMin	737	16	16	11	9	14	25	29	22	15	34	40	28		
	PrecStd	146	40	34	32	32	29	41	33	39	33	42	37	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	10.8	13.2	18.3	22.0	24.9	26.3	28.1	21.4	17.6	13.4	12.2		
		MCWB	9.5	9.0	10.0	13.1	15.4	17.7	17.2	17.4	16.4	14.4	12.1	11.0		
	2%	DB	9.3	9.7	11.0	15.4	19.6	22.2	24.2	24.0	18.7	15.5	12.4	11.3		
		MCWB	8.2	8.2	9.0	11.5	14.3	16.0	17.2	17.2	14.8	13.1	11.4	10.4		
	5%	DB	8.3	8.7	9.9	13.1	17.2	20.2	22.1	21.2	16.8	14.3	11.2	9.8		
		MCWB	7.2	7.4	8.4	10.0	12.7	15.4	16.3	16.1	13.5	12.4	10.2	9.1		
	10%	DB	7.3	7.7	8.8	11.4	15.0	18.2	20.2	19.1	15.7	13.3	10.3	8.8		
		MCWB	6.3	6.5	7.3	8.8	11.5	14.2	15.5	15.2	13.0	11.9	9.4	8.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.6	10.6	13.5	16.3	18.8	19.0	18.7	16.8	15.0	12.7	11.4		
		MCDB	10.3	10.4	12.4	16.8	20.9	21.4	23.2	24.0	20.7	16.8	13.0	12.0		
	2%	WB	8.5	8.6	9.6	11.7	14.6	17.1	17.9	17.7	15.5	13.8	11.5	10.5		
		MCDB	9.2	9.4	10.7	15.0	18.7	21.4	22.7	22.4	17.6	15.0	12.1	11.2		
	5%	WB	7.4	7.7	8.5	10.5	13.2	15.7	17.0	16.7	14.5	12.9	10.4	9.2		
		MCDB	8.1	8.6	9.5	12.6	16.5	19.3	20.7	20.6	16.2	13.8	11.1	9.7		
	10%	WB	6.4	6.6	7.6	9.3	12.0	14.5	16.1	15.7	13.5	12.1	9.5	8.0		
		MCDB	7.2	7.5	8.6	11.0	14.7	17.8	19.4	18.7	15.2	13.0	10.2	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.9	5.6	6.7	7.7	7.6	7.6	7.8	6.9	5.8	4.9	4.1		
		MCDBR	5.9	5.7	6.6	8.6	10.8	11.5	11.7	11.3	8.4	6.9	5.0	4.8		
	5% WB	MCWBR	5.6	5.0	5.0	5.2	6.4	6.6	6.0	5.5	5.3	5.1	4.6	4.8		
		MCWBR	5.7	5.7	5.9	8.3	9.5	9.9	9.9	9.8	7.5	5.7	5.0	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.330	0.360	0.385	0.388	0.387	0.387	0.373	0.358	0.340	0.316	0.304		
		taud	2.350	2.397	2.316	2.256	2.285	2.312	2.314	2.373	2.419	2.445	2.408	2.324		
	Ebn at Noon		662	776	821	844	860	864	857	849	818	754	659	597		
		Edh at Noon	59	76	101	121	123	122	119	106	90	72	56	51		
(44) All-Sky Solar Radiation	RadAvg		0.69	1.36	2.47	3.81	4.78	5.04	4.79	3.94	2.90	1.64	0.84	0.54		
	RadStd		0.06	0.15	0.26	0.41	0.32	0.49	0.50	0.27	0.24	0.14	0.09	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page